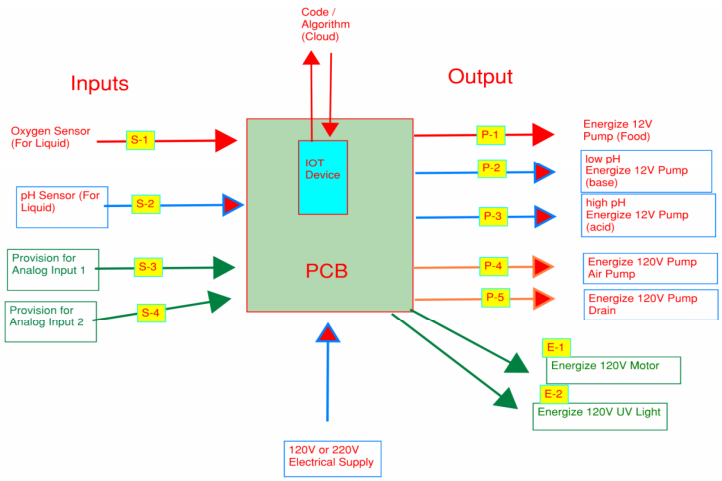
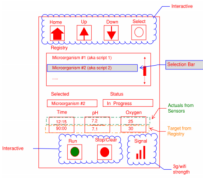


Inputs		Outputs		
S-1	Oxygen Sensor	P-1		12V Pump (Food Source)
S-2	pH Sensor	P-2		12V Pump (Base)
S-3	(Future)	P-3		12V Pump (Acid)
S-4	(Future)	P-4		120V Pump (Air)
		P-5		120V Pump (Drain)
		E-1		120V Motor
		E-2		120V UV Light

CODE BASE												
	Duration	Input Frequency	Oxygen	pH		SENSOR CYCLE			SCRIPT CYCLE			BUTTON
Parameters/Variables >	(Minutes)	(Minutes)	(PPM)									
Registry	Y	Z	O	P		P-1	P-2	P-3	P-4	E-1	E-2	P-5
Script 1	120	10	5	7		S-1 > O	S-2 < P	S-2 > P	ON	ON	OFF	
Script 2 (Future)	"--"	"--"	"--"	"--"		S-1 > O	S-2 < P	S-2 > P				
Script 3 (Future)	"--"	"--"	"--"	"--"		S-1 > O	S-2 < P	S-2 > P				

PERFORMANCE RUN

Operator Engagement







Operator Navigates Registry Script interacting with the Controller. (Simple Programmable Screen with Button Interface)

Operator Selects '**RUN**' (Button Interface)

Script is followed per Parameters/Variables

	PARAMETERS / VARIABLES				SENSOR CYCLE			SCRIPT CYCLE			BUTTON
	Duration	Input Frequency	Oxygen	pH	P-1	P-2	P-3	P-4	E-1	E-2	P-5
Script Conditions >	Y	Z	O	P				ON/OFF	ON/OFF	ON/OFF	Button
Selected Script "Script 1"	120	10	5	7	S-1 > O	S-2 < P	S-2 > P	ON	ON	OFF	ON/OFF

Script is followed for Duration Y

Input Frequency Z

Every 'Z' Minutes Run the Following

Y

10 Minutes

Input S-1 = S-1
Input S-2 = S-2

		ENERGIZE	Duration
IF	S-1 > O	P-1	5s
IF	S-2 < P	P-2	5s
IF	S-2 > P	P-3	5s

ON	ON	OFF

ON/OFF

Script Complete

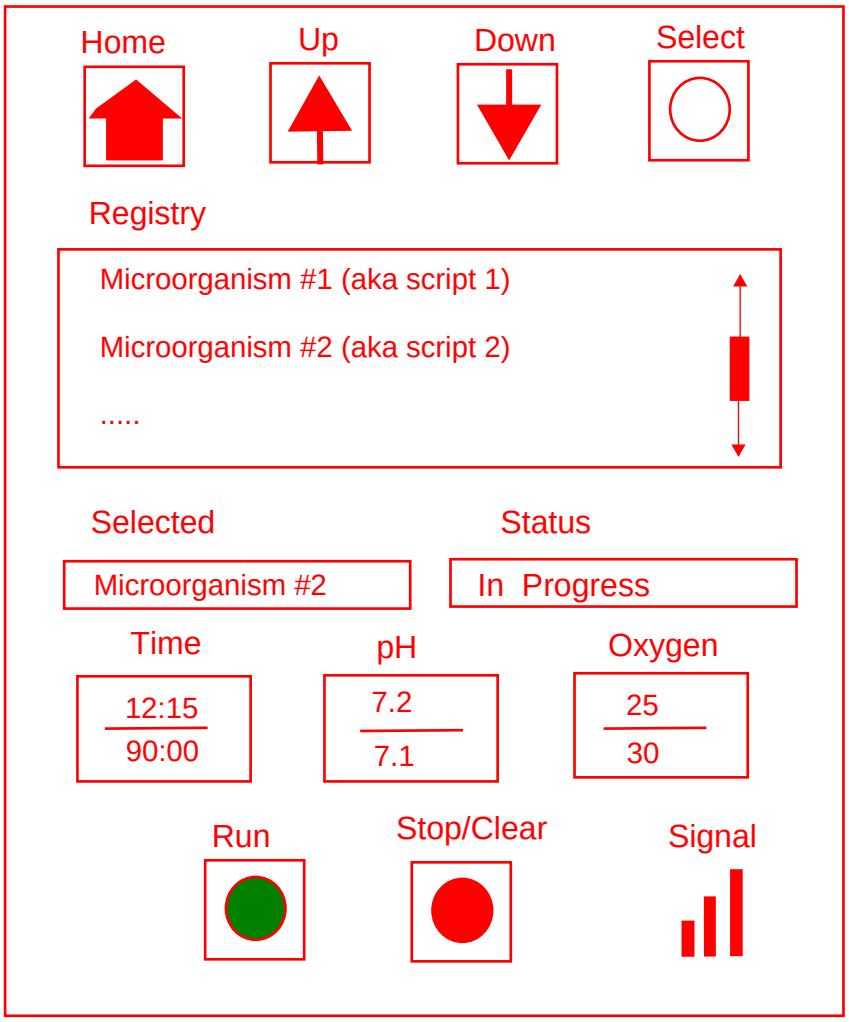
P-5
Button
ON/OFF

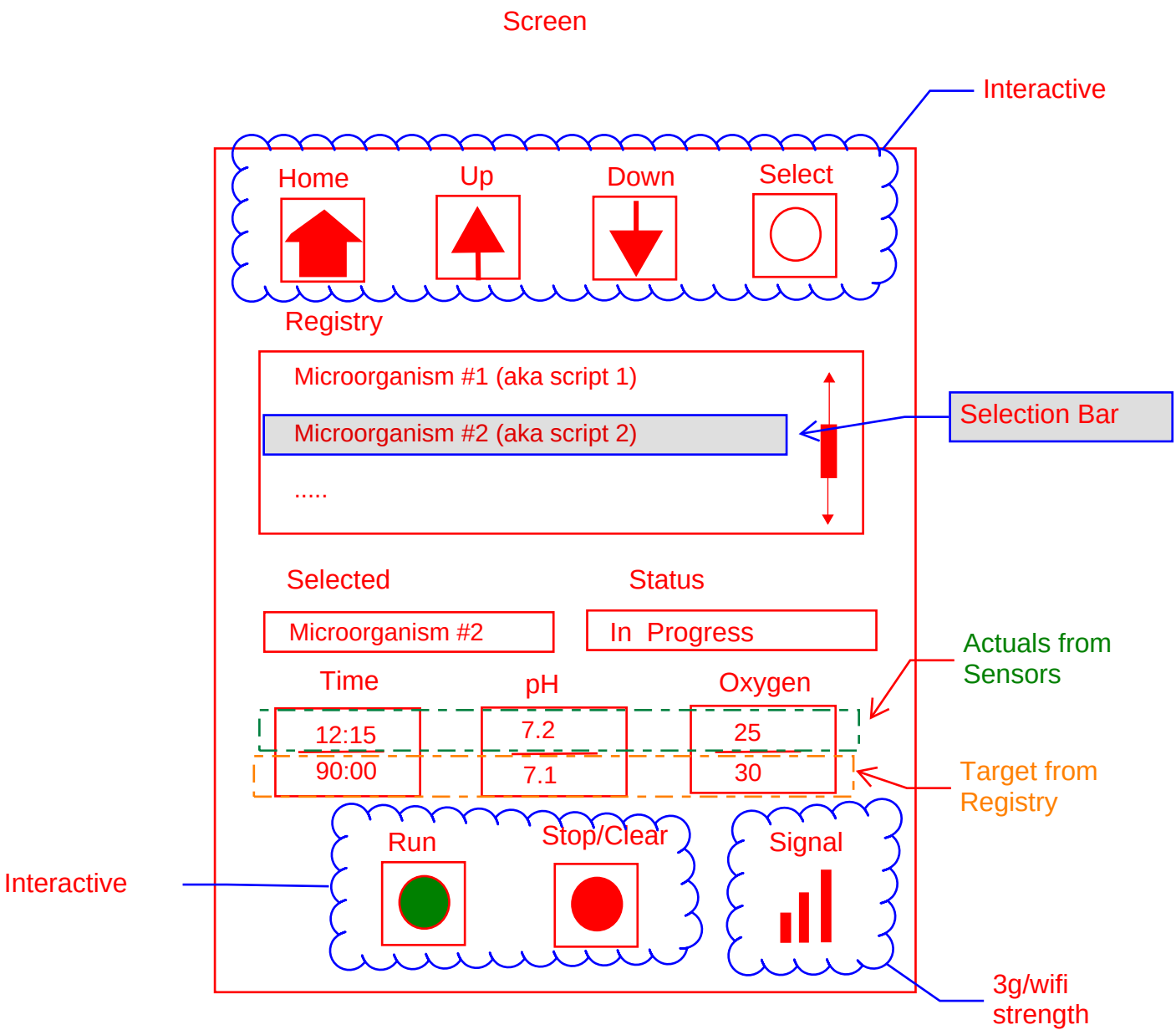


Operator Select '**END**' to clear cache

Drain

Screen


Home
Up
Down
Select
Registry
Microorganism #1 (aka script 1)
Microorganism #2 (aka script 2)
.....
Selected
Status
Microorganism #2
In Progress
Time
12:15
90:00
pH
7.2
7.1
Oxygen
25
30
Run
Stop/Clear
Signal


Screen
Interactive
Home
Up
Down
Select
Registry
Microorganism #1 (aka script 1)
Microorganism #2 (aka script 2)
.....
Selection Bar
Selected
Status
Microorganism #2
In Progress
Actuals from Sensors
Time
pH
Oxygen
12:15
7.2
25
90:00
7.1
30
Target from Registry
Run
Stop/Clear
Signal
Interactive
3g/wifi strength

Controller Functionality / IOT

1) Database/Registry

Client can purchase Scripts and Expand Registry

(Need to be able to push new scripts to the device or their account.)

Registry

Name	pH	Oxygen	Duration
Microorganism #1 (aka script 1)	7.2	32	90:00
Microorganism #2 (aka script 2)	8.5	22	120:00

.....

2) Monitoring/Historical Reports

Client gets billed for each 'Run'

Need to be able to monitor how many times the device is used